Nishant Porwal

nishantporwal203@gmail.com | +91 62324 42003 | LinkedIn | GitHub | LeetCode

Education

National Institute of Technology Delhi *Graduating 2027*
B.Tech in Computer Science and Engineering CGPA: 7.8

Work Experience

RESEARCH INTERN | Davise Lab, NIT Delhi June 2026 – July 2026

- Designed **TerraAlert**, an AI-powered landslide monitoring and disaster response framework leveraging geospatial segmentation, predictive modeling, automated rerouting and intelligent alert dissemination for emergency decision support.
- Applied DeepLabV3+ with a ResNet-50 backbone for landslide detection and severity classification using georeferenced satellite imagery, achieving a 20% improvement in affected-zone localization accuracy.
- Engineered a genetic-algorithm driven route optimization system for real-time, hazard-aware evacuation planning.

WEB DEVELOPER | NIT Delhi June 2025 – Dec 2025

- Migrated legacy website (serving **20k+ monthly visitors)** from WordPress to Next.js.
- Built **5+ internal web applications**, including faculty profile systems used by **150+ faculty members**, improving administrative workflows and overall usability.
- Developed and maintained full-stack applications using Next.js, Node.js, MongoDB, REST APIs, Git, and CI/CD automation.

Technical Projects

PeerLink

- Built a real-time full-stack video communication platform with **React, TypeScript, Node.js, WebRTC and WebSockets**.
- Engineered low-latency audio/video communication using **native WebRTC** and custom **WebSocket signaling**, handling **SDP negotiation**, **ICE candidate buffering**, concurrent peer sessions and automated socket cleanup.
- Implemented stateless **JWT authentication** with HTTP-only cookies and **Zod validation**, used **Recoil** for reactive call-state management, and containerized frontend/backend services with **Docker** for consistent deployment.

Legal AI Assistant: Model Serving & Deployment

- Deployed a fine-tuned **Gemma-3B** model as a scalable REST API on **GCP Vertex AI**, serving clause-level classification requests with an **85% F1-score** on a custom Indian legal dataset.
- Built a multi-layer QnA inference pipeline using a **Longformer** model (ROUGE 0.68) and optimized serving latency via **batched inference** & **model quantization**, reducing manual review time by an estimated 75%.
- Designed a **Flask** backend exposing AI features as structured REST endpoints, managing model I/O, request preprocessing, and production deployment on **GCP Cloud Run.**

IT Support RAG

- Built a Retrieval-Augmented Generation (RAG) system using **Python, FastAPI, Qdrant, SentenceTransformers, and LlamaIndex** for document ingestion and semantic search.
- Developed a modular pipeline with configurable chunking, collection reuse, prompt customization and REST APIs for scalable document retrieval.
- Implemented streaming responses and containerized the application using **Docker** for reproducible deployment and efficient local development.

Technical Skills

Languages: Python, C++, C, JavaScript, SQL
Frameworks & Libraries: Nodejs, Flask, RestAPI, TensorFlow, scikit-learn
Core Concepts: Data Structures & Algorithms (DSA), Machine Learning, Deep Learning, Object-Oriented Programming (OOPs), DBMS (Database Management and System), Operating System (OS)
Tools & Technologies: MongoDb, Git, GitHub, Docker, MySQL, Github Copilot, Claude

Leadership & Activities

- Senior Coordinator & Design Head: Training and Placement Cell, NIT Delhi || Coordinator of Technical Club, UpVision.
- Editorial Board Member of Institutional Newsletter of the National Institute of Technology, Delhi
- Member of the Organizing Committee for the ICCSDT International Conference (two consecutive years).